

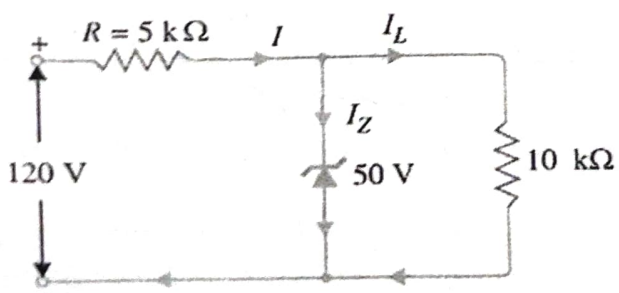
(Please write your Enrollment Number)

Enrollment No. _____

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech, Semester: 1st Semester
(December, 2024)

Subject Code: BEC-103	Subject: Electronics Workshop
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1	(2.5*8=20)	CO Mapping
a) What is the difference between Zener Breakdown and avalanche Breakdown.		CO1
b) What is an Electrodynamometer type Wattmeter?		CO2
c) What is an Amplified DC Meter?		CO3
d) Give the relationship between α , β and γ of a transistor		CO1
e) Briefly explain the True-RMS responding voltmeter?		CO3
f) Discuss about the cut-off and saturation mode of operations of BJT.		CO2
g) Simplify using Boolean Algebra $Y = AB + \overline{AC} + A\overline{B}C (AB + C)$		CO4
h) Convert the following expression in canonical POS form. $Y = (A + B) (\overline{B} + C)$		CO4
UNIT I		CO Mapping
Q2	Demonstrate and justify the input and output characteristics of common base configuration of BJT.	(10) CO1
Q3	Describe the construction, characteristics, and operation of a Zener diode. Explain how it can be used as a voltage regulator. For the circuit shown below, find (i) the output voltage (ii) the voltage drop across series resistance (iii) the current through zener diode. 	(10) CO1
UNIT II		CO Mapping
Q4	Explain the working of emitter bias circuit and derive the expression of its stability factor S.	(10) CO2

Q5	With a neat diagram explain in detail the construction and principle of PMMC Instruments. What are shunts and multipliers? Derive the expression for both, with reference to meters used in electrical circuits.	(10)	CO2
UNIT III			CO Mapping
Q6	With the help of block diagram explain the working principle of Electronic Voltmeter?	(10)	CO3
Q7	(a) Draw the block diagram of Digital Frequency meter? (b) Explain the principle of operation of Vector Voltmeter?	(5+5)	CO3
UNIT IV			CO Mapping
Q8	Use K-Map to minimize and express the function in SOP form. Implement the expression using NAND gates only. $F = \sum m(0, 1, 2, 6, 7, 10, 12, 15) + \sum d(3, 8, 13, 14)$	(10)	CO4
Q9	(a) Implement the following Boolean functions using a 3:8 decoder. Use block diagram for the decoder. $F1(A, B, C) = \sum m(2, 4, 7)$ $F2(A, B, C) = \sum m(0, 3)$ $F3(A, B, C) = \sum m(0, 2, 3, 4, 7)$ (b) Differentiate between i. Positive Logic and Negative Logic ii. Multiplexer and Demultiplexer	(6+4)	CO4